Function

A Function is a block of code which only runs when it is called.

Function avoid code repetition

uses of function

1.code reusability

2.code organization

3.reduce code duplication

4.easy to debugging

5.improve readability

In [ ]:
```
#simple example
```

In [5]:
```
def add():
    a=10
    b=20
    c=a+b
    print(c)
add()
```
```
30
```

In [7]:
```
def add(a,b):
    c=a+b
    print(c)
add(5,5)
```
```
10
```

In [9]:
```
def add():
    a=10
    b=5
    print(a+b)
add()
```
```
15
```

1.operators

In [12]:
```
def add(a,b):
    print(a+b)
def sub(a,b):
    print(a-b)
add(5,5)
sub(5,5)
```
```
10
0
```

2.Global variable

In [15]:
```
a=10
b=20
def add():
    print(a+b)
add()
```
```
30
```

In [19]:
```
# chnage variable value in global
a=10
b=20
def add():
    global a
    a=5
    print(a+b)
add()
```
```
25
```

In [21]:
```
# global variable access any where in the code
a=10
b=5
def add():
    print(a+b)
add()
print(a)
```
```
15
10
```

3.Local variable

In [11]:
```
def add():
    a=10
    b=5
    print(a+b)
add()
```
```
15
```

In [1]:
```
# local varible access inside function only
def add():
    a=10
    b=5
add()
print(a)
```
```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[1], line 5
      3     b=5
      4 add()
----> 5 print(a)

NameError: name 'a' is not defined
```

Arguments

Arguments are values passed to a function when calling it

In [5]:
```
def name(data):
    print("Hello",data)
name("Reeta")
```
```
Hello Reeta
```

In [7]:
```
def name(data):
    print("hello",data)
name("Reeta")
name("Hashini")
name("subi")
name("lux")
```
```
hello Reeta
hello Hashini
hello subi
hello lux
```

In [23]:
```
def data(name,age,city):
    print("Name: ",name,",","Age: ",age,",","City: ",city)
data("Reeta",20,"Salem")
```
```
Name:  Reeta , Age:  20 , City:  Salem
```

defalut parameter

In [29]:
```
def data(name="Unknown"):
    print("Hello",name)
data("Reeta")
data()
```
```
Hello Reeta
Hello Unknown
```

keyword argument

In [34]:
```
def data(name,age):
    print("I am",name)
    print("My name is",name,"and I am",age,"old.")
data(name="Reeta",age=20)
```
```
I am Reeta
My name is Reeta and I am 20 old.
```

Positional Arguments

In [37]:
```
def data(name,age):
    print("I am",name)
    print("My name is",name,"and I am",age,"old.")
data("Reeta",20)
```
```
I am Reeta
My name is Reeta and I am 20 old.
```

Python *args and* *kwargs

1.*args

*args is used to pass multiple arguments to a function without specifying the number of arguments.

In [61]:
```
def num(*name):
    print("hello",name[0])
num("Reeta","Mano","Preethi")
```
```
hello Reeta
```

In [63]:
```
def data(*age):
    print("my age is",age[4])
data(23,45,4,34,78,49)
```
```
my age is 78
```

2.**kwargs

**kwargs is used to pass multiple keyword arguments to a function.

In [67]:
```
def data(**d):
    print(d)
data(name="reeta",age=20,ciy="Salem")
```
```
{'name': 'reeta', 'age': 20, 'ciy': 'Salem'}
```

Scope

Scope defines where a variable can be accessed in a program.

1.local scope

A variable declared inside a function. It can be accessed only inside that function.

In [72]:
```
def num():
    a=10
    print(a)
num()
```
```
10
```

In [74]:
```
def num():
    a=10
num()
print(a)
```
```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[74], line 4
      2     a=10
      3 num()
----> 4 print(a)

NameError: name 'a' is not defined
```

In [ ]:
```
# function inside another function
```

In [76]:
```
def num():
    a=10
    def n():
        print(a)
    n()
num()
```
```
10
```

2.global scope

A variable declared outside a function. It can be accessed both inside and outside functions.

In [82]:
```
a=10
def num():
    print(a)
num()
```
```
10
```

In [84]:
```
a=10
def num():
    print(a)
num()
print(a)
```
```
10
10
```

In [86]:
```
a=10
def num():
    print(a)
    def n():
        print(a)
    n()
num()
```
```
10
10
```

In [ ]:
```
# change global variable
```

In [88]:
```
a=10
def num():
    global a
    a=20
num()
print(a)
```
```
20
```

3.Nonlocal Keyword

The nonlocal keyword is used to work with variables inside nested functions.

In [94]:
```
def num():
    a=10
    def num1():
        nonlocal a
        a=20
    num1()
    print(a)
num()
```
```
20
```

Practic Question

1.Write a function to add two numbers.

In [98]:
```
def add():
    a=5
    b=5
    print(a+b)
add()
```
```
10
```

In [100…
```
def add(a,b):
    print(a+b)
add(5,5)
```
```
10
```

2.Write a function to find the factorial of a number

In [109…
```
def num(n):
    f = 1
    for i in range(1, n+1):
        f *= i
    return f

print(num(5))
```
```
120
```

3.Write a function to check whether a number is even or odd.

In [4]:
```
def eo():
    a=10
    if a%2==0:
        print("even")
    else:
        print("odd")
eo()
```
```
even
```

4.Write a function to check whether a number is prime or not.

In [7]:
```
def prime():
    a=3
    if a>1:
        if a%2==0 and a!=2:
            print("not prime")
        else:
            print("prime")
    else:
        print("ot prime")
prime()
```
```
prime
```

5.Write a function to find the largest of three numbers.

In [10]:
```
a=10
b=20
c=30
def large():
    if a>b and a>c:
        print("A")
    elif b>a and b>c:
        print("B")
    else:
        print("C")
large()

```
```
C
```

task

create calculator

In [83]:
```
num1=int(input("enter num1:"))
num2=int(input("enter num2:"))
opt=input("enter opt:")
def add():
    a=num1 + num2
    print("add:",a)
def sub():
    a=num1 - num2
    print("sub:",a)
def multi():
    a=num1 * num2
    print("multi:",a)
def div():
    a=num1 / num2
    print("div:",a)

if opt=="+":
    add()
elif opt=="-":
    sub()
elif opt=="*":
    multi()
elif opt=="/":
    div()
```

```python
else:
    print("invalied")
```

```
add: 15
```